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DEVIN AEBI

SKILLS

Python, C++, ROOT, Data Analysis, Visualization, Tensorflow, SQL, Big Data, Machine Learning, Deep Neural Networks, Monte Carlo Simulation, Git, Teaching, Leading, Management

EXPERIENCE

Compact Muon Solenoid Experiment — CERN Large Hadron Collider, Geneva Switzerland — *Research Assistant*

SEPTEMBER 2022 - APRIL 2026

- Leader of integration and coordination of the GEM-CSC integrated trigger. Managed international team of graduate students, post-docs, and research scientists. Produced several public detector notes.
- Worked on the first ME0 Detector stack for CMS. Tested electronic components, analyzed early test beam data, and helped organize design and logistics.
- Level 3 Trigger Contact in GEM Detector Performance Group. Lead contact for trigger related issues in the GEM-CSC subsystem for CMS.
- Shifts as CSC Detector On Call, responsible for accurate and consistent data taking during CMS Run 3 data from 2022-2026.

Texas A&M University, College Station TX — *Research Assistant*

SEPTEMBER 2018 - PRESENT

- Developed GEM Subsystem Software Alignment, a novel data-driven approach to aligning the GEM subsystem to a resolution of 100 microns. Produced first alignment for the new detector system for the start of Run 3 (2022).
- Worked on the CSC Optical Trigger Motherboard algorithms, creating proper L1 trigger objects at a collision rate of 25 nano seconds.
- Analyzing CMS Run 3 data searching for a new particle X->HH->bbWW. An analysis over 62 fb-1 of proton-proton collision data and hundreds of terabytes of simulation. Trained and applied several deep neural network models to produce the most significant possible constraints on the production cross section.

EDUCATION

Texas A&M University — *Ph.D - Physics*

AUGUST 2022 - PRESENT, COLLEGE STATION TX

Texas A&M University — *M.Science - Physics*

SEPTEMBER 2018 - AUGUST 2022, COLLEGE STATION TX

University of Michigan — *B.Science - Physics, B.Mus - Performance*

SEPTEMBER 2014 - MAY 2018, ANN ARBOR MI